Lab Class: Physics
Moment of inertia and angular acceleration

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1 Introduction

This lab experiment demonstrates the fundamental laws of angular acceleration and the moment of inertia. It is part of a university course and serves as a learning tool. It specifically has the goal to demonstrate the relationship between the different physical quantities in an angular movement.

2 Experimental Setup

The experiment was set up in a laboratory provided by the *Hochschule München University of Applied Sciences*, which is used for a variety of experiments on classical mechanics (room, G247).
For the measurements a rotatably mounted system, consisting of three differently sized aluminum discs was used. The welded-together discs have a rod (length: $830 \pm 0.5mm$) attached frontally at its center. The rod is equipped with two movable barbell masses ($m = 308.7 \pm 0.2g$ each). The distance between the two was measured using a ruler (uncertainty: $\pm 0.2mm$). In addition, every disc has a groove to hold the pulling thread, which can be rolled up around any disc of choice. To make the system work, a mass piece ($m_g = 40.9 \pm 0.5g$) is attached to the end of the thread.
The discs are held in place by a locking mechanism with a lever. When released, the mass piece m_g is pulled down by the gravitational force ($F = m_g \cdot g \approx 0.40N$) and accelerates at a constant rate. The acceleration acts tangentially on the disc and therefore causes an equally constant angular acceleration α in the rotatably mounted system. A defined torque τ is generated, the magnitude of which is dependent on the disc size chosen.
A marking was placed on both the discs and the wall behind them, placed to aligned with each other. This way the number of rotations n could be measured by waiting for the markings to align again. Using this method and a built in stopwatch on a smartphone (model: *Xiaomi 11*), the duration of a certain amount of rotations was measured, flagging the time t after each rotation. The time measurement was carried out by the same person counting the number of rotations n, therefore a human uncertainty factor ($\pm 0.2s$) was considered for the time measurement.
All data collected, was immediately recorded in a *JupyterLab* file. All calculations and graphical representations of the experiment were conducted within the same file.
All measurements were executed ten times and averaged to improve accuracy, resulting in varying uncertainties of time.

3 Experimental Produce

The measurements were executed in two variations and were used to showcase three different physical relationships, which are as follows.

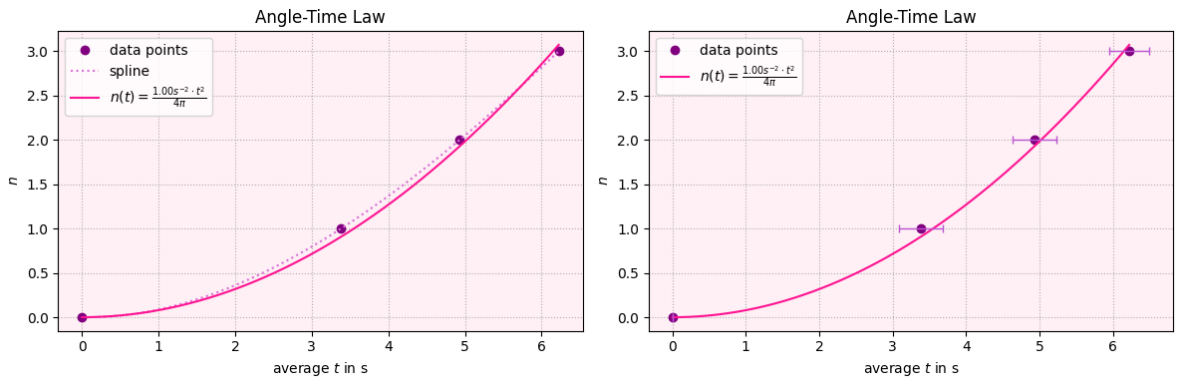
a. Angle-Time-Law

For the first set of measurements, the largest of the three discs (diameter: $173.1 \pm 0.2mm$) was used, the diameter of the distance between the barbell masses was fixed ($d_{m_1} = 375 \pm 0.2mm$) and the time for three rotations ($n = 3$) was measured.

t	$n = 0$	$n = 1$	$n = 2$	$n = 3$	$n = 4$
meas 1	0	6.34	9.24	11.3	13.25
meas 2	0	6.26	8.84	11.26	13.11
meas 3	0	6.44	9.2	11.5	13.35
meas 4	0	6.61	9.38	11.67	13.52
meas 5	0	6.38	9.1	11.28	13.14
meas 6	0	6.22	8.98	11.11	12.99
meas 7	0	6.33	8.99	11.14	13.03
meas 8	0	6.28	9.04	11.24	13.03
meas 9	0	6.22	8.93	11.09	13.02
meas 10	0	6.24	9.04	11.16	13.09
average <t>	0	6.33	9.07	11.28	13.15

(1)

The time that passed after each rotation could simply be read off of the collected data.
The data suggests a quadratic growth, which fits the known formula $n = \frac{\alpha}{4\pi} \cdot t^2$, as shown in the figure below.



(2)

Due to the constant property of the angular acceleration, it's magnitude ($\alpha_1 = 1.03 \pm 0.036 \frac{rad}{s^2}$) can be calculated.

$$\alpha = \frac{n \cdot 4\pi}{t^2_{(n)}} \quad \text{or more accurately:} \quad \alpha = \left(\sum_{i=0}^{n} \frac{n_i \cdot 4\pi}{t^2_{(i)}}\right) \cdot \frac{1}{n}$$

For improved accuracy, the plot shown in the figure was generated with a curve fit. The barely divergent data points suggest a measurement accurate to theory.
The time uncertainties (figure (2), right), consist of the human uncertainty factor ($\pm 0.2s$) and the standard error from averaging the multiple measurements (± 0, ± 0.095, ± 0.099, ± 0.073) added together.

b. Second Newtonian Law for Rotational Motion

For the second and third set of measurements two smaller discs (diameters: $112.6 \pm 0.2mm$ and $52.7 \pm 0.2mm$) were used. The measurements were executed for four rotations ($n = 4$) in the same manner as the first set (tabel (1)).

t	$n=0$	$n=1$	$n=2$	$n=3$	$n=4$
meas 1	0	6.34	9.24	11.3	13.25
meas 2	0	6.26	8.84	11.26	13.11
meas 3	0	6.44	9.2	11.5	13.35
meas 4	0	6.61	9.38	11.67	13.52
meas 5	0	6.38	9.1	11.28	13.14
meas 6	0	6.22	8.98	11.11	12.99
meas 7	0	6.33	8.99	11.14	13.03
meas 8	0	6.28	9.04	11.24	13.03
meas 9	0	6.22	8.93	11.09	13.02
meas 10	0	6.24	9.04	11.16	13.09
average <t>	0	6.33	9.07	11.28	13.15

diameter: 112.6 ± 0.2mm

t	$n=0$	$n=1$	$n=2$	$n=3$	$n=4$
meas 1	0	4.19	6.05	7.57	8.94
meas 2	0	4.22	6.2	7.58	8.97
meas 3	0	4.38	6.37	7.81	9
meas 4	0	4.54	6.43	7.89	9.44
meas 5	0	4.53	6.43	7.9	9.2
meas 6	0	4.38	6.4	7.89	9.21
meas 7	0	4.55	6.44	7.94	9.31
meas 8	0	4.25	6.22	7.79	9.07
meas 9	0	4.55	6.45	7.94	9.28
meas 10	0	4.64	6.48	8.01	9.31
average <t>	0	4.42	6.35	7.83	9.17

diameter: 52.7± 0.2mm

(3)

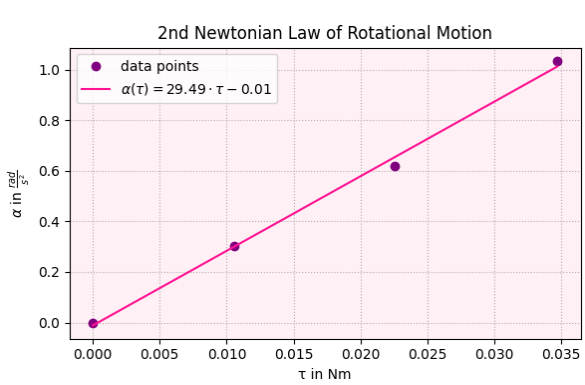
The differing diameters d of the discs, including the first measurement (tabel (1)), resulted in different torque values τ.
The torque ($\tau_1 = 0.035 \pm 0.0005Nm$, $\tau_2 = 0.023 \pm 0.0004Nm$, $\tau_3 = 0.011 \pm 0.0002Nm$) for each disc was calculated, using the force that acted on the system ($\approx 0.40N$).

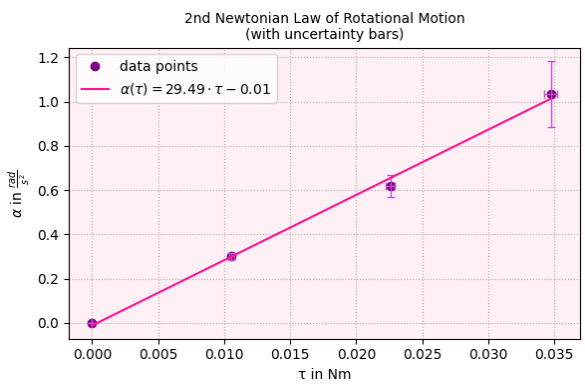
$$\tau = m_g \cdot g \cdot \frac{d}{2}$$

The angular acceleration can be obtained the same way as for the first measurements ($\alpha_1 = 1.03 \pm 0.036\frac{rad}{s^2}$, $\alpha_2 = 0.62 \pm 0.010\frac{rad}{s^2}$, $\alpha_3 = 0.30 \pm 0.005\frac{rad}{s^2}$).
For each set of measurements the calculation was done for each number of rotations n (with $n > 0$) and averaged, to improve accuracy.
This reveals a relationship of the torque τ and the angular acceleration α the factor which is the moment of inertia I, applying Newtons second law to rotational motion.

$$\alpha = \frac{\tau}{I}$$

This linear relationship is visualized in the figure below.





(4)

To obtain the values for the plot shown in the figure, the torque τ of each disc and their corresponding angular acceleration α were used as parameters in a linear regression.

The data points, relatively consistent with the linear regression, confirm the linear nature of their relationship. Due to minor measurement inaccuracies, the intercept of the linear regression is not exactly zero.
To calculate the uncertainties (figure (4), right), all uncertainties were added to or subtracted from the values in every formula used to calculate the torque τ and α in such a manner that the highest possible value for each could be obtained. Then the original values of τ and α were subtracted from their maximum values. A growth of the uncertainties, relative to the original values, can be observed.

c. Dependence of Moment of Inertia on the Distance of the Barbell Masses

For the last three sets of measurements, only one disc (diameter: $112.6 \pm 0.2mm$) was used to keep the torque constant ($\tau = 0.023 \pm 0.0004Nm$). The distance between the two barbell masses was adjusted. First the second measurement (tabel (2), left) was used (without $n = 0$) as a reference, then the distance was increased from the original ($d_{m_2} = 548 \pm 0.2mm$) and lastly narrowed down ($d_{m_3} = 300 \pm 0.2mm$), to examine the effect on the moment of inertia I. The time was measured again for four rotations ($n = 4$) in the same manner as for the other measurements.

t	$n=1$	$n=2$	$n=3$	$n=4$
meas 1	5.53	8.12	10	11.67
meas 2	5.68	8.07	9.97	11.56
meas 3	5.58	8.04	10.02	11.7
meas 4	5.65	8.12	10.15	11.85
meas 5	5.69	8.21	10.18	11.75
meas 6	5.38	7.87	9.73	11.42
meas 7	5.43	7.9	9.9	11.52
meas 8	5.54	8.09	10.07	11.64
meas 9	5.28	7.72	9.58	11.33
meas 10	5.34	7.68	9.71	11.31
average <t>	5.51	7.98	9.93	11.58

mass distance: 548 ± 2mm

t	$n=1$	$n=2$	$n=3$	$n=4$
meas 1	3.1	4.5	5.73	6.66
meas 2	3.1	4.89	6.1	7.17
meas 3	3.27	4.76	5.9	6.92
meas 4	3.25	4.73	5.94	6.95
meas 5	3.58	4.89	6.16	7.16
meas 6	3.24	4.79	5.95	7
meas 7	3.15	4.6	5.81	6.9
meas 8	3.55	4.98	6.23	7.21
meas 9	3.21	4.64	5.79	6.78
meas 10	3.11	4.59	5.75	6.72
average <t>	3.26	4.74	5.94	6.95

mass distance: 300 ± 2mm

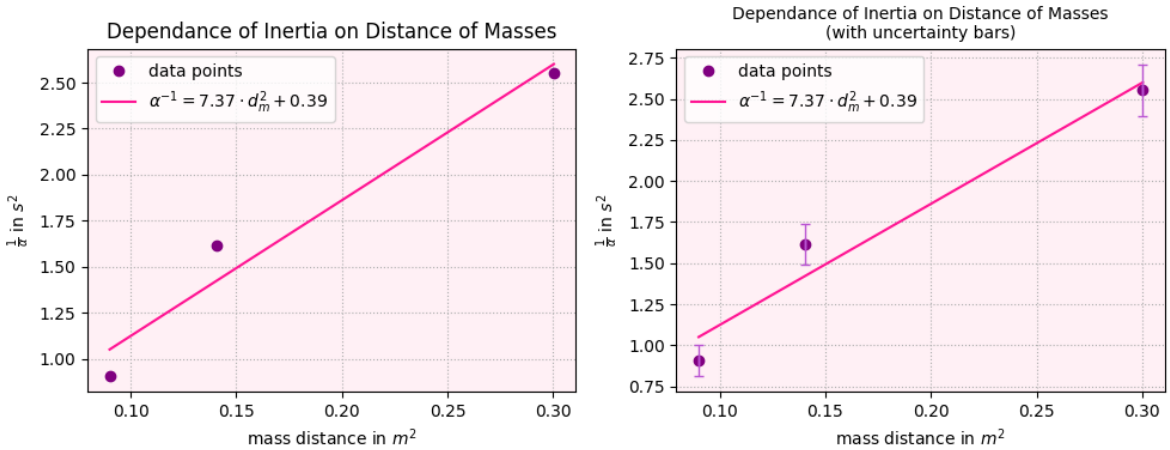
(5)

The data shows a clear change in angular acceleration ($\alpha_1 = 1.10 \pm 0.032 \frac{rad}{s^2}$, $\alpha_2 = 0.62 \pm 0.010 \frac{rad}{s^2}$, $\alpha_3 = 0.39 \pm 0.009 \frac{rad}{s^2}$) for every change of the mass distance d_m. As the torque is constant, the relationship of torque τ and angular acceleration α by a factor of the moment of inertia I, suggests a change in the moment of inertia.

$$\alpha_{(I)} = \frac{\tau}{I}$$

When inversing the formula for the angular acceleration ($\frac{1}{\alpha} = \frac{I}{\tau}$), the constant property of the torque τ, enables a clear recognition of the linear looking relationship, as shown in the figure

below.



(6)

The plot shown in the figure, was obtained by using the mass distance squared d_m^2 and the inverse angular acceleration $\frac{1}{\alpha}$ as parameters in a linear regression.
The intercept of the linear regression is not zero, as the moment of inertia I consist not only of the moment of inertia of the masses ($I_{masses} = \frac{1}{2}m \cdot d_m^2$), but also the constant moment of inertia of the discs and rod.

$$I_{system} = I_{mass} + I_{disc-rod}$$

The data points are sufficiently, but not overly consistent with the linear regression. That was to be expected, as treating the masses m as point masses, simplifies reality in favor of easier calculations. For a more accurate interpretation, the masses cannot be treated as point masses and the data would have to be curve fitted.
To calculate the uncertainties of the plot (figure (6), right), the same method was used for d_m^2 and $\frac{1}{\alpha}$ as for the parameters of figure (4). Only a very slight growth of the uncertainties can be observed.

4 Summary

The results confirmed the expected physical relationships, predicted by theory. Minor discrepancies between theory and experimental values were observed. Due to imprecise methods of measuring, the uncertainty values of the data were improvable.
Despite that, the experiment successfully demonstrated the fundamental principles of the moment of inertia and angular acceleration.

Literature

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